

Dino v0.0.1 — CPU-Free 8-Bit Sequencer

Objective

Design a manual, CPU-free 8-bit sequencer that:

- Generates a 16-bit address stream using chained SN74LS590N counters and a 555 timer clock.
- Loads program bytes into SRAM via an 8-bit DIP switch in LOAD mode.
- Uses a 74LS645 octal bus transceiver to arbitrate the shared data bus, preventing contention between DIP switch, SRAM, and LED display.
- Switches cleanly between LOAD and RUN modes via SPDT-controlled DIR and /OE signals.
- Reads and displays stored data sequentially on an LED bank in RUN mode.

Bill of Materials

Qty	Component
2	SN74LS590N 8-bit binary counters
1	MCM60256AP-10 32KB SRAM
1	555 Timer (clock generation)
1	SPDT switch (LOAD/RUN mode)
2	Pushbuttons (WRITE, CLOCK)
1	8-bit DIP switch (LOAD mode data input)
8	LEDs (RUN mode output)
8	10 k Ω resistors (pull-down for DIP switch)
8	220–470 Ω resistors (LED current limiting)
1	74LS645N octal bus transceiver
1	10 k Ω Resistor (R1)
1	100 k Ω Resistor (R2)
1	1 μ F (C1)
1	0.1 μ F (C2)
—	Breadboard, jumper wires, 5 V power supply

Bus Architecture

Device	Bus Role	Mode
DIP Switch	Drives bus	LOAD
SRAM	Drives & reads	LOAD & RUN

Device	Bus Role	Mode
LED Display	Reads from bus	RUN

74LS645 Control Signals

Mode	/OE	DIR	Function
LOAD	Low	High	DIP → SRAM write
RUN	Low	Low	SRAM → LED read
IDLE	High	Low	Bus tri-stated

Design Considerations

- **Power-on Reset:** Include a reset mechanism (pushbutton or RC network) on SN74LS590 counters to initialize the address to 0 on startup.
- **Decoupling:** Place 0.1 μ F bypass capacitors at each VCC-GND pin of the SRAM, 74LS645, and SN74LS590 devices.
- **Control-Line Biasing:** Add pull-ups or pull-downs on 74LS645 DIR and /OE so the bus has a defined state if the SPDT switch is between positions.
- **Default IDLE State:** IDLE mode tri-states the bus; the DIP switch will float (use pull-downs) and LEDs remain off.
- **Resistor Placement:** All LED current-limit resistors go on the anode side; ensure DIP switch inputs have pull-downs.
- **Timing Constraints:** Keep the 555-timer frequency well below the 74LS645 maximum toggle rate (~ 25 MHz) and within the SRAM's access time. We will generate ~ 6.8 Hz at around 52.7% Duty Cycle with the provided C1, R1 & R2
- **Diagram Reference:** Consider adding a simple block diagram showing 74LS645 side-A to DIP, side-B to SRAM/LED bus.
- **Terminology:** Use "IDLE" consistently for tri-state rather than mixing "FLOAT".

Edge-Case Checks

- **Global Reset & SRAM /CE:** Ensure SN74LS590 and 74LS645 power up in a known state (use RC reset or pushbutton) and SRAM's /CE is tied low with outputs truly tri-stating on /OE high.
- **Debounce Write/Clock Buttons:** Add RC or Schmitt-trigger debouncing to prevent mechanical bounce causing extra writes or steps.
- **Synchronous Timing:** Verify 555 clock edges meet SN74LS590 setup/hold times; ensure duty-cycle respects LS family timing even at low frequencies.
- **Glitch Immunity on Mode Switch:** Use a two-pole switch or RC delay so /OE changes only after DIR settles, avoiding transient bus contention.
- **Biasing Between Modes:** Confirm DIP switch side inputs have proper pull-downs so they don't float when /OE is high.

- **Power Supply Decoupling:** Place 0.1 μF bypass caps at every IC's VCC/GND pins and consider a 4.7 μF bulk cap on the 5 V rail.
- **Mode-by-Mode Verification:** Test LOAD and RUN separately – verify bus integrity in each and decide if the IDLE tri-state is needed in practice.

Build Steps

1. Program Counter

1. Chain two SN74LS590N counters for a 16-bit address.
2. Connect the 555 timer to the lower counter's CP0.
3. Chain the lower's ripple carry (RC) to the upper's CP0.
4. Route Q0–Q15 $\rightarrow$ SRAM A0–A15.

2. Bus Transceiver Integration

1. Place the 74LS645 between DIP switch and shared data bus:
 - Side A $\leftarrow$ DIP switch lines
 - Side B $\leftarrow$ SRAM D0–D7 and LED bank
2. Drive DIR: High in LOAD, Low in RUN
3. Drive /OE: Low to enable, High to tri-state (IDLE)

3. Mode Switch Logic

Use the SPDT switch to set:

- **LOAD:** Enable transceiver, DIR = High, SRAM write enabled
- **RUN:** Enable transceiver, DIR = Low, SRAM output enabled
- **IDLE (optional):** /OE = High for all bus devices

4. Manual Programming (LOAD Mode)

1. Set mode to LOAD
2. Set DIP switch to desired byte
3. Press WRITE (SRAM /WE low) to store
4. Press CLOCK to increment address
5. Repeat for each byte

5. Program Execution (RUN Mode)

1. Set mode to RUN
2. Start the 555 timer (or step manually)
3. LEDs display memory contents sequentially

Completion Criteria

- No bus contention in LOAD or RUN modes

- DIP switch drives only in LOAD
- SRAM drives only in RUN
- LEDs read bus correctly
- Bus lines tri-stated when idle